OTT Streaming Content Trend Analysis

Netflix catalog: genre, ratings, release trends and regional content mix

Exploratory data analysis and content strategy report | Veda Technology Internship, Project 1 | October 2026

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Executive summary

This report analyses 8,807 titles from Netflix's catalog (6,131 movies and 2,676 TV shows, snapshot through September 2021) to understand how the content mix has changed and what that suggests for future content decisions. The data was cleaned and explored in Python (Pandas, Seaborn, Plotly) and the core aggregations were repeated in SQL (SQLite). Three findings stand out:

- **Movies are getting shorter.** Movies under 90 minutes rose from 16.8% (released 2000-09) to 34.7% (2016-21), and average runtime fell from 112 to 95 minutes.
- **Growth is coming from international markets.** Spain (+172%), Mexico (+159%), Japan (+115%) and South Korea (+88%) grew fastest between 2012-16 and 2017-21, mostly from small bases.
- **Most TV shows are one-and-done.** 67% of TV shows have a single season and only 17% have three or more.

Metric	Value
Total titles	8,807
Movies / TV shows	6,131 (69.6%) / 2,676 (30.4%)
Release years covered	1925 to 2021
Titles added to Netflix	2008 to Sept 2021
Adult-rated titles (TV-MA, R, NC-17)	45.5%
Median movie length	98 minutes
United States share of titles	41.9%
India share of titles	11.9%

1. Objective and approach

The aim of the project is to go from a raw catalog dump to a short narrative about what has changed in the catalog and why it might matter to the business, the kind of analysis a content strategy or growth team would run. The work follows five steps: load and inspect the data, clean it, explore it visually, validate the core aggregations in SQL, and turn the strongest patterns into three actionable recommendations.

2. Dataset and cleaning

The dataset is the public *Netflix Movies and TV Shows* catalog from Kaggle (8,807 rows, 12 columns: type, title, director, cast, country, date added, release year, rating, duration, genres and description). It describes the catalog only; it contains no viewership or engagement data. There are no duplicate rows or duplicate title IDs.

Issue found	Treatment
3 rows (Louis C.K. specials) have the runtime stored in the rating column ('74 min', '84 min', '66 min') and an empty duration	Moved the value into duration and set rating to missing before any other filling
Missing director (2,634), cast (825), country (831)	Filled with 'Unknown'; Unknown countries are excluded from country analysis

Issue found	Treatment
Missing rating (4 + the 3 fixed rows)	Labelled 'Not Rated' instead of guessing a value
Missing date added (10)	Left empty; year added is only used where available
Duration mixes minutes (movies) and seasons (TV shows)	Split into two numeric columns: duration_min and seasons
Genres and countries are comma-separated lists	Split and expanded so each genre or country is counted once per title
Ratings are many overlapping codes	Grouped into age groups: Kids, Older Kids, Teens, Adults, Not Rated

The cleaned data was saved and loaded into SQLite, with two helper tables (title_genres and title_countries) so that multi-valued genres and countries could be joined correctly.

3. Exploratory analysis

3.1 Catalog composition and growth

Movies make up 69.6% of the catalog and TV shows 30.4%. Content additions accelerated sharply after 2015: Netflix added 429 titles in 2016, 2,016 in 2019 (the peak) and 1,879 in 2020, with 1,498 in the first nine months of 2021. TV shows have accounted for between 25% and 41% of yearly additions since 2016, so the movie-heavy mix has stayed fairly stable.

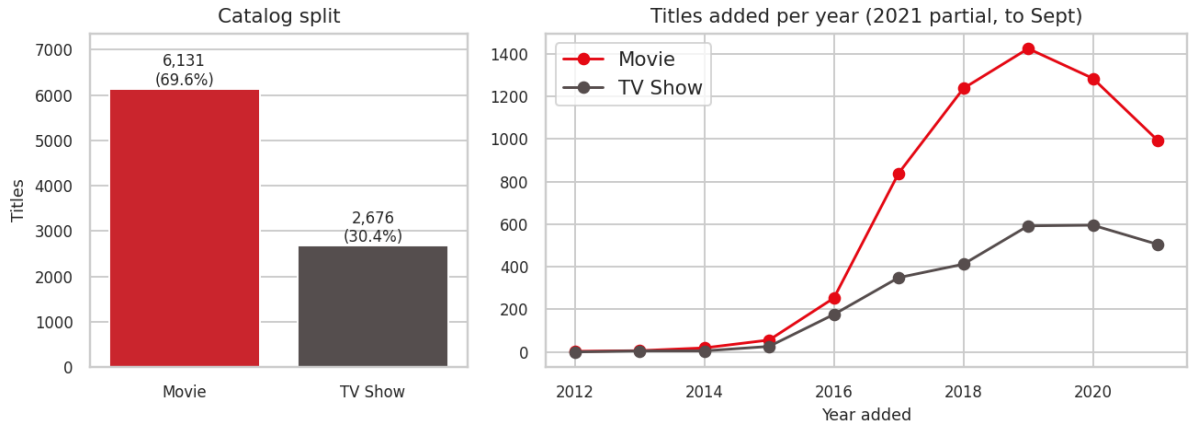


Figure 1. Movies vs TV shows in the catalog, and titles added per year by type.

3.2 Genres

Dramas (2,427 tags) and Comedies (1,674) dominate, followed by Documentaries, Action & Adventure and TV Dramas. The 'International Movies' and 'International TV Shows' tags appear on 47% of titles; they describe origin rather than genre, so they were excluded from this chart.

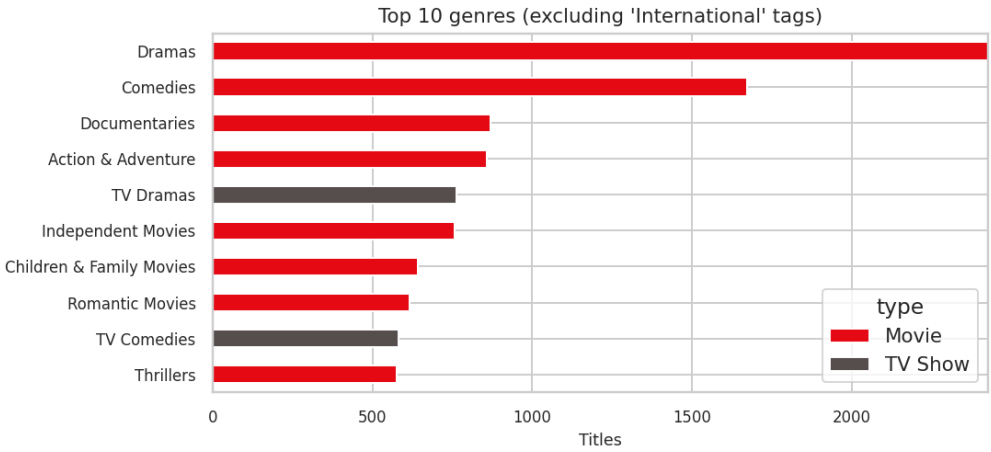


Figure 2. Top 10 genres by number of titles, split by movies and TV shows.

3.3 How the genre mix has changed

Looking at titles by release period, the biggest genres are taking a smaller slice: Dramas fell from 17.8% to 10.9% of genre tags, Comedies from 15.7% to 7.7% and Action & Adventure from 8.1% to 3.1%, which points to a catalog spreading across a wider range of niches. TV Dramas moved the other way, rising from 3.4% to 7.9%, and Documentaries jumped from 2.1% in 2008-12 to 7.4% in 2013-16.

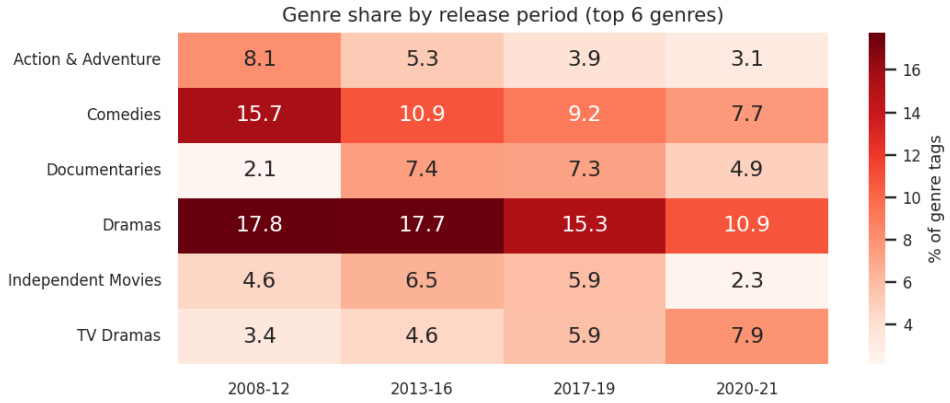


Figure 3. Share of genre tags for the six largest genres, by release period.

3.4 Regional content mix

The United States appears on 41.9% of titles (3,690 titles including co-productions), followed by India (1,046), the United Kingdom (806), Canada (445) and France (393). About 9.4% of titles have no country listed. Releases from every major country peak around 2018-19; the lower counts for 2020-21 should be read with caution because the snapshot ends in September 2021.

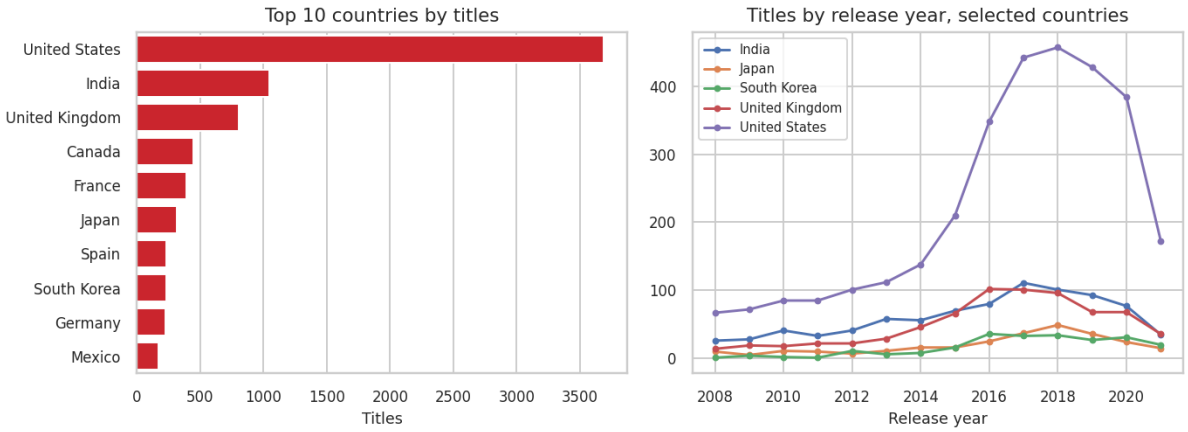


Figure 4. Top 10 content-producing countries, and titles by release year for selected countries.

3.5 Ratings and audience

Mature content leads: TV-MA (3,207) and TV-14 (2,160) are the two most common ratings, and 45.5% of all titles are adult-rated. Children's content is a larger part of TV than of movies: Kids and Older Kids titles are 29.5% of TV shows against 20.7% of movies.

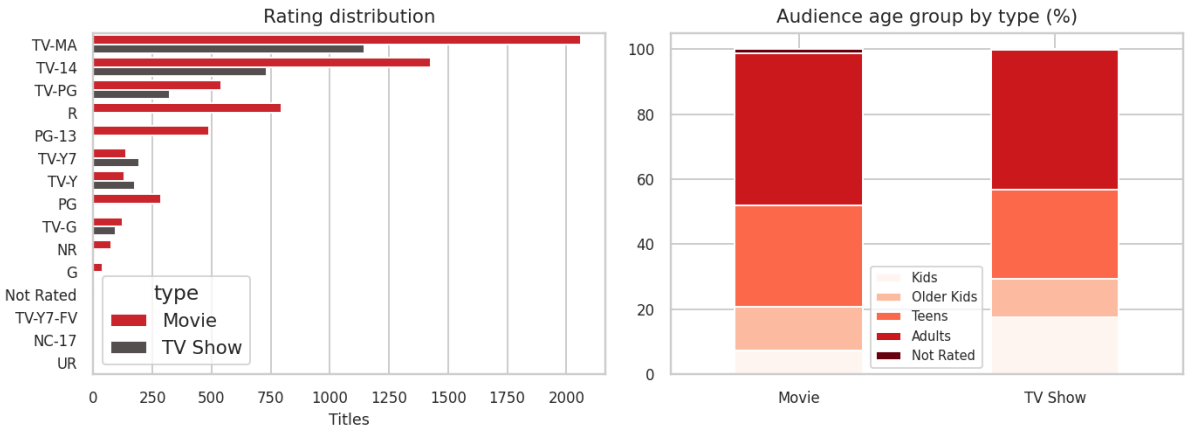


Figure 5. Rating distribution and audience age group by content type.

3.6 Movie duration

The median movie length is 98 minutes. For movies released since 2000, the yearly median fell from 107 minutes in 2000 to 95 minutes in 2019, and the share of movies under 90 minutes doubled across the three periods shown in Section 5.

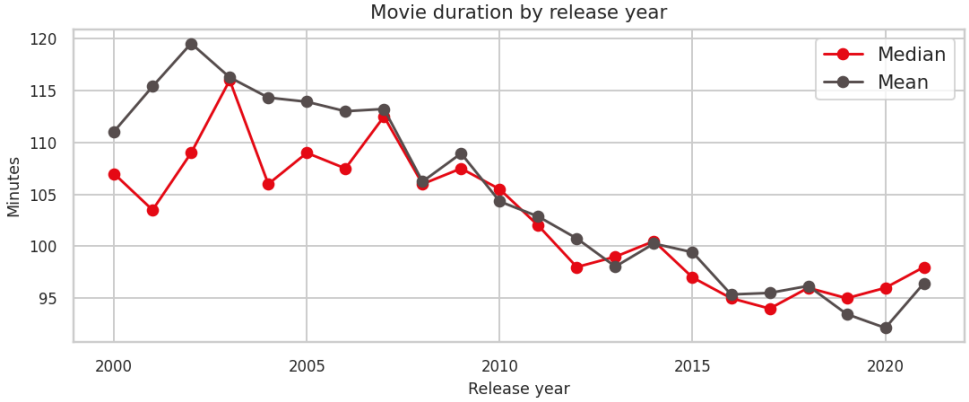


Figure 6. Median and mean movie duration by release year (2000 onwards).

3.7 TV show seasons

67.0% of TV shows have a single season, 15.9% have two, 7.4% have three and 9.7% have four or more. The pattern is not just caused by new shows waiting for renewal: among shows released in 2018 or earlier, 68.5% still have only one season.

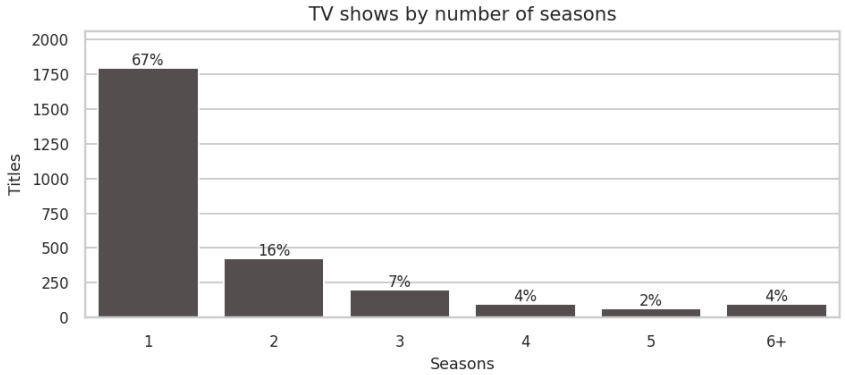


Figure 7. Number of TV shows by season count.

4. SQL analysis

Ten queries were written to cross-check the Python results and to practise aggregations, joins and window functions. All ten reproduce the figures reported above.

Query	Question answered	Technique
Q1	Movies vs TV shows split	GROUP BY, subquery
Q2	Titles added per year, with TV share	Conditional aggregation
Q3	Top 10 genres	Helper table, GROUP BY
Q4	Top 10 countries	Helper table, GROUP BY
Q5	Top genres among Indian titles	Three-table JOIN
Q6	Movie duration by period	CASE, AVG
Q7	Age-group share within each type	Window function (SUM OVER)
Q8	Seasons distribution	CASE, subquery
Q9	Top genre per country	JOIN, ROW_NUMBER()
Q10	Fastest-growing countries	JOIN, HAVING, NULLIF

Titles added per year (Q2)

Year added	Movies	TV shows	Total	TV share %
2015	56	26	82	31.7
2016	253	176	429	41.0
2017	839	349	1188	29.4
2018	1237	412	1649	25.0
2019	1424	592	2016	29.4
2020	1284	595	1879	31.7
2021	993	505	1498	33.7

Fastest-growing countries by release period (Q10)

Country	2012-16	2017-21	Growth %
Spain	58	158	172.0
Mexico	41	106	159.0
China	44	97	120.0
Japan	75	161	115.0
United States	909	1884	107.0
South Korea	77	145	88.0

Most common genre in five major markets (Q9)

Country	Top genre	Titles
India	Dramas	662
Japan	Anime Series	143
South Korea	Korean TV Shows	132
United Kingdom	British TV Shows	225
United States	Dramas	835

5. Content strategy recommendations

To: Content Acquisition Team **Subject:** What to greenlight next

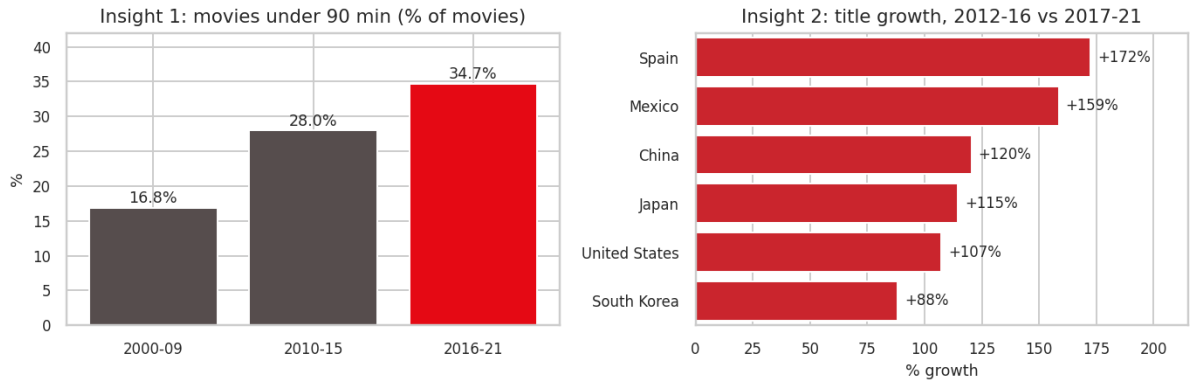


Figure 8. Evidence for Insights 1 and 2.

Insight 1: movies are getting shorter

Evidence: the share of movies under 90 minutes rose from 16.8% to 28.0% to 34.7% across 2000-09, 2010-15 and 2016-21, while average runtime fell from 112 to 95 minutes. **Recommendation:** prioritise tightly paced films in the 85-100 minute range. Shorter films are likely cheaper to make and easier to fit into one evening, but the catalog cannot confirm viewer preference, so test the idea with a small slate first.

Insight 2: growth is coming from international markets

Evidence: comparing titles released in 2012-16 with 2017-21, Spain grew 172% (58 to 158 titles), Mexico 159%, Japan 115% and South Korea 88%. The United States still makes up about 42% of the catalog and its share was roughly flat (38.9% to 39.6%). India is the largest non-US source (11.9% of titles) but grew more slowly (+37%). **Recommendation:** increase investment in Spanish-language, Mexican, Korean and Japanese originals, where growth is strongest and the catalog is still small. For India, where volume is already high, focus on fewer, higher-quality titles rather than more of them.

Insight 3: most TV shows are one-and-done

Evidence: 67.0% of TV shows have one season and only 17.1% have three or more (Figure 7); the share is still 68.5% for shows released in 2018 or earlier. **Recommendation:** treat the limited series with a planned ending as the default format and budget for it that way. Commit to multi-season runs only for shows that have proven themselves, since long-running series are the exception in this catalog.

6. Limitations

- The dataset describes the catalog, not viewership, so the recommendations are hypotheses about what to greenlight, not proof of what audiences prefer.
- The snapshot ends in September 2021, so figures for 2021 are partial.
- Growth is measured by release year, not by the year a title was added to Netflix.
- Co-productions are counted once for each country involved, so country totals add up to more than the number of titles.
- A one-season show may have been cancelled or designed as a limited series; the data cannot tell these apart.
- Around 9% of titles have no country listed and are excluded from regional analysis.

7. Conclusion and next steps

The catalog has grown quickly and is becoming broader: more niches, more international titles, shorter movies and TV shows that often end after one season. Together these patterns suggest a strategy built on shorter films, selective bets on fast-growing international markets, and limited series as the default TV format. With viewership or engagement data, the next step would be to test whether shorter movies and international titles actually perform better, and to compare Netflix with Amazon Prime's catalog.

Tools used: Python (Pandas, NumPy, Matplotlib, Seaborn, Plotly), SQL (SQLite) and Google Colab. Dataset: Netflix Movies and TV Shows, Kaggle (shivamb).